



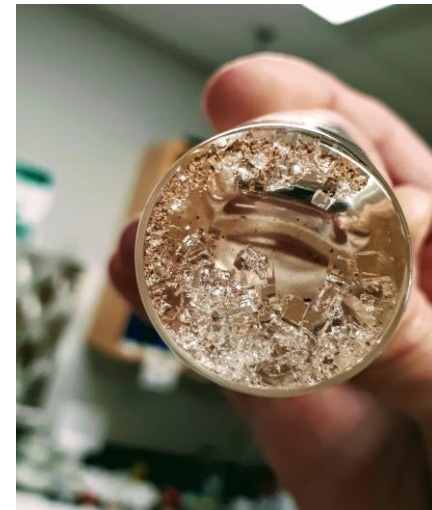
Welcome to the 17th Canadian Chemical Crystallography Workshop (CCCW26)

Introduction to Crystals, Crystallography, and Crystal Growth

MAY 19, 2026

What is a crystal?

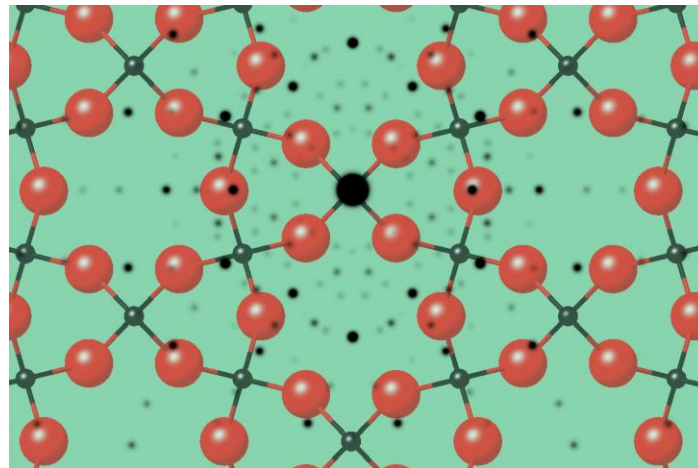
- Historically, crystals were first defined in terms of their external morphologies, with the angular and symmetry relationships between their faces.
- This led to a definition based on the three-dimensional translational periodicity inferred from the periodicity of the array of Bragg peaks in the diffraction patterns of most crystals.
- More recently crystals have been discovered for which the array of Bragg peaks is not periodic in three dimensions so that the repeating atomic arrangement lacks translational periodicity in at least one direction.
- The Nobel Prize in Chemistry 2011 was awarded to Dan Shechtman "for the discovery of quasicrystals"



What is a crystal?

Direct-space definition:

- A solid is a crystal if its atoms, ions and/or molecules form, on average, a long-range ordered arrangement.
- In most crystals the arrangement is a periodic array that is governed by the rules of translational symmetry. In aperiodic crystals (incommensurate and quasicrystals) the arrangement is not periodic in three dimensions but is nevertheless still fully ordered, where the ordering follows particular mathematical rules.



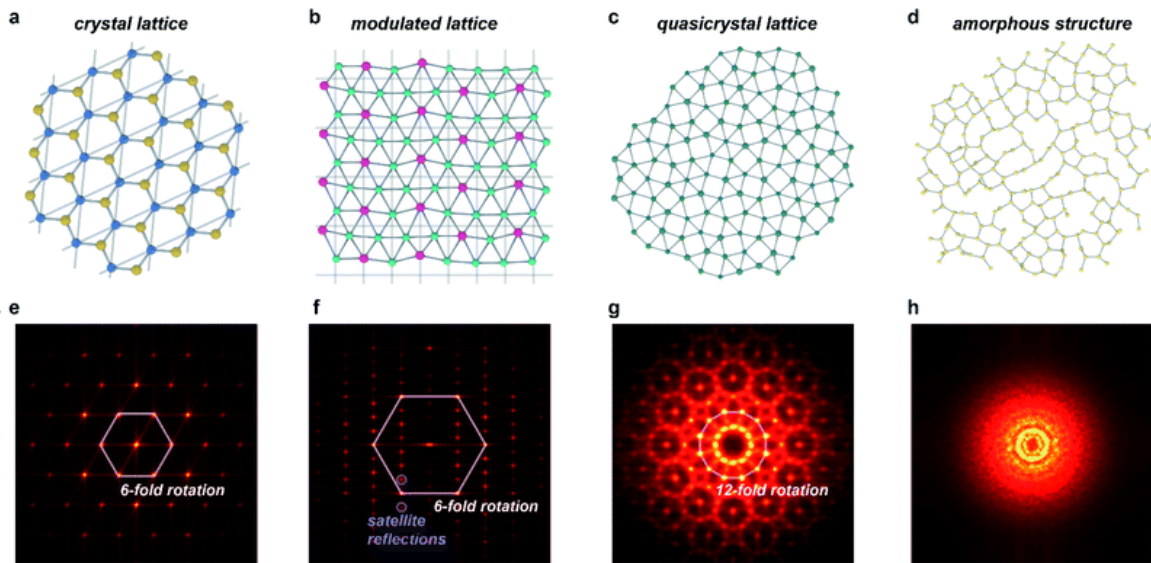
Carolyn P. Brock. Change to the definition of “crystal” in the IUCr Online Dictionary of Crystallography. *IUCr Newsletter*. **2021**, 29(2). <https://www.iucr.org/news/newsletter/volume-29/number-2/change-to-the-definition-of-crystal-in-the-iucr-online-dictionary-of-crystallography>

What is a crystal?

4

Reciprocal-space definition:

- A material is a crystal if it has essentially a sharp diffraction pattern.
- The word essentially means that most of the intensity of the diffraction is concentrated in relatively sharp Bragg peaks, besides the always present diffuse scattering.



Order in aperiodic crystallography and its manifestation in diffraction. In Oppenheim, J.J.; Skorupskii, G.; Dincă, M. *Aperiodic metal–organic frameworks*. *Chem. Sci.*, **2020**, *11*, 11094–11103.

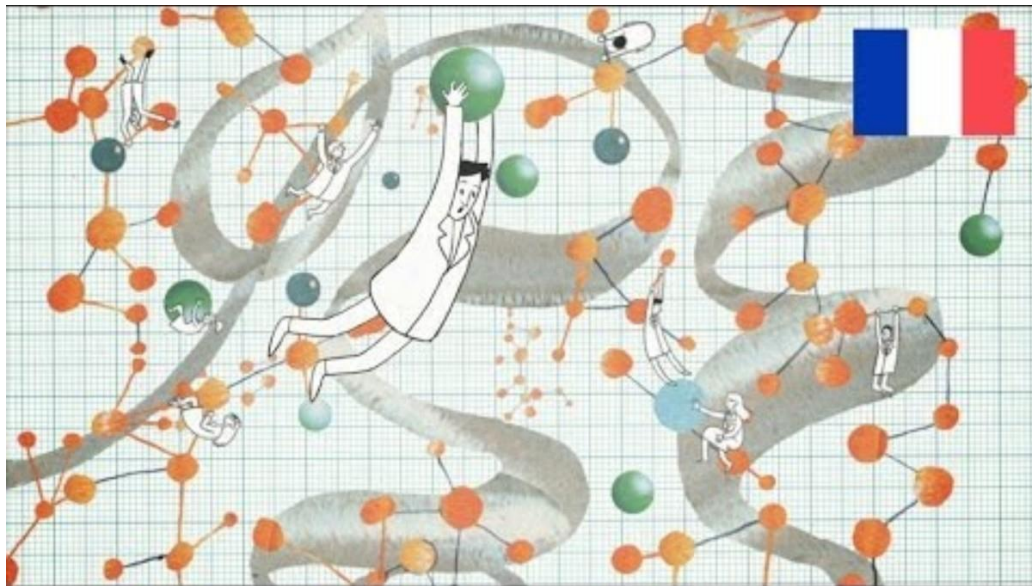
What is a crystallography?

5

“Crystallography is the science that develops our abilities to extract structural information using crystallographic techniques.”

Sven Lidén, former President of the International Union for Crystallography

<https://www.iucr.org/news/newsletter/volume-26/number-1/what-is-crystallography>



Why does it matter?

[The disaster I never imagined having to worry about](#)

30-minute YouTube Video from Veritasium on polymorphism and polymorphs of the HIV drug, Ritonavir.

How do I grow a crystal?

6



With guidance from Dr. Paul Boyle!